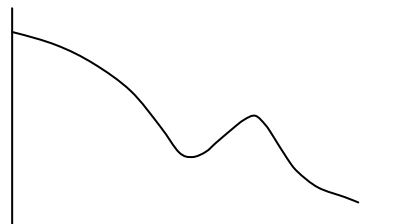


Question	Marking guidance	Mark	Comments
02.1	Substitutes into equation correctly, condoning omission of powers of ten ✓ Correctly calculates 2 values of k including correct powers of ten ✓ Calculates all 3 values correctly and finds the mean of 1.49×10^{-45} (m^3) ✓	3	1.57×10^{-45} ; 1.47×10^{-45} ; 1.42×10^{-45}
02.2	Finds the percentage uncertainty using the half range of candidate's values from 02.1 ecf ✓	1	Expect to see 5% Accept $\frac{\text{Max}-\text{mean}}{\text{mean}} \times 100$ or $\frac{\text{Mean}-\text{min}}{\text{mean}} \times 100$
02.3	Calculation approach – calculates 2 densities correctly ✓ Calculates all 3 densities correctly and concludes that they are sufficiently similar ✓ Algebraic approach: uses $m = 1.67 \times 10^{-27}$ A and $V = \frac{4}{3}\pi r^3$ to shown that density is a function of $\frac{A}{R^3}$ ✓ $\frac{A}{R^3} = \frac{1}{k}$ which is a constant ✓	2	Values of density: $2.5(4) \times 10^{17}$; $2.7(1) \times 10^{17}$; $2.8(1) \times 10^{17}$; Ignoring powers of ten: MAX 1 Expect to see $\rho = \frac{3 \times 1.67 \times 10^{-27}}{4\pi} \cdot \frac{1}{k}$.

Question	Marking guidance	Mark	Comments
02.4	MAX 3 Top path: Little or no deviation ✓ Middle path: Reflected back along own path ✓ Deviation >> deviation of top path ✓ Distance of closest approach is on middle alpha particle path ✓	3	Accept deviation of up to 60° by eye Accept parallel to original path if gap is less than 2 mm Expect between 20° and 160° (by eye) but compared with deviation of top path Must not touch nucleus No marks awarded for paths with sharp angle turns (apart from head on)
02.5	Correct general shape ✓ ... with non-zero first minimum labeled θ ✓	2	Allow subsequent, maxima and minima MP2 dependant on MP1 

Question	Marking guidance	Mark	Comments
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	20	C
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	21	C
--	----	---

	22	D
--	----	---

	23	c
--	----	---

	24	B
--	----	---

	25	c
--	----	---

	26	B
--	----	---

	27	c
--	----	---

	28	A
--	----	---

	29	c
--	----	---

	30	c
--	----	---

	31	B
--	----	---

Question	Marking guidance	Additional comments/guidelines	Mark
05.1	Top line: 236 and 90 ✓ Bottom line: 92 and 37 ✓	First MP = both proton numbers correct Second MP = both nucleon numbers correct	2
05.2	Attempts to find a mass defect OR attempts to use $E = mc^2$ ✓ 0.31337×10^{-27} (kg) ✓ $2.8(2) \times 10^{-11}$ (J) ✓	Mass defect must be to at least 2 sf to get mark Correct to 2 sf	3
05.3	Converts to J eg 3.2×10^{-13} seen OR use of $E_k = \frac{1}{2}mv^2$ ✓ 1.95×10^7 (m s ⁻¹) ✓	Seen to at least 3 sf Accept 1.96 for 1.951	2
05.4	Recognisable conservation of momentum equation seen ✓ Speed of neutron = 6.6×10^6 (m s ⁻¹) ✓	Accept calculations using mass of hydrogen = 1 and mass of deuterium = 2	2
05.5	MAX 2 Most / all collisions (between neutron and deuteron) will not be head on (so less momentum will be lost by the neutron) ✓ There will be a range of initial speeds ✓ Deuterium (nucleus) may not be stationary ✓		2
Total			11

Question	Key
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Question	Key
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	21	D
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	22	B
--	----	---

	23	D
--	----	---

	25	C
--	----	---

	26	C
--	----	---

	27	C
--	----	---

	28	D
--	----	---

	29	C
--	----	---

	30	A

Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	154 (neutrons) seen ✓ Attempts to calculate a mass defect (expect 2.02 u) OR attempts to convert a mass in u to kg ✓ 3.4×10^{-27} (kg) ✓	May be seen as (252 – 98) $(154 \text{ n} + 98 \text{ p}) = 254.04862 \text{ u} = 4.219 \times 10^{-25} \text{ kg}$ $m_{\text{cf}} = 4.186 \times 10^{-25} \text{ kg}$ Allow 3.3×10^{-27} (kg)	3	AO1 AO1 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Calculates difference in mass, including 4 neutrons ✓ Converts mass difference to energy (u to eV, or kg to J) ✓ 210 (MeV) ✓	Expect 0.22532 u or $3.7 \times 10^{-28} \text{ kg}$	3	AO2 AO2 AO2
Total			6	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Volume is proportional to temperature OR reference to Charles' Law OR statement of ideal gas law with N/n , p constant ✓ Length is proportional to volume as cross-sectional area is constant ✓	alternative: the KE/speed of the gas particles increases ✓ collisions between particles and container walls/liquid column remains the same (so length increases) ✓	2	2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	extrapolation to the x -intercept leading to -250 to -270 ($^{\circ}\text{C}$) ✓	allow mathematical extrapolation no more than 3sf	1	1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Read off l for chosen T (not $l=0$) ✓ Conversion of $^{\circ}\text{C}$ to K ✓ Use of $pV = nRT$ and $V = lA$ ✓ 0.75 to 0.80 (mm^2) ✓	alternative: two read-offs (in order to calculate gradient) OR uses gradient calculated in 5.2 if applicable ✓ conversion of $^{\circ}\text{C}$ to K ✓ calculates gradient AND $A = nR/(P \times \text{gradient})$ ✓ 0.75 to 0.80 (mm^2) ✓ allow POT errors for MP1,3	4	4 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	MAX 2: Uncertainty due to extrapolation from a narrow range ✓ Large percentage uncertainty in measuring the length of the air column because the length is small ✓ Difficulty in measuring length of air column described (eg parallax, ruler in the water bath)✓	Accept idea that the air column and the water bath are at different temperatures when temperature is measured (eg slow heat transfer through glass) ✓ Accept idea of meniscus (affecting l) Condone the pressure in the laboratory varies Condone evaporation of liquid	2	2 × AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
05.5	Greater gradient AND greater y -intercept OR (line would extrapolate to) same x -intercept ✓ double gradient AND double y -intercept ✓	line must extend into the positive x - region	2	2 × AO4
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Marking guidance	Mark	Comments
01.1	80.2 or 80.3 or 80.4 (°C)	1	Only accept these values.

Question	Marking guidance	Mark	Comments
01.2	Determines freezing time ✓ Use of $E = Pt$ OR $Q = mL$ ✓ Allow 0.022 to 0.025 (kg) ✓	3	Allow 7 to 8 (mins) For MP2: If time used is within range 6.0 to 8.5, condone use in minutes. If time is outside this range, t must be used seconds. For MP2: condone use of 7.8 for Q. 2 sf answer only.

Question	Marking guidance	Mark	Comments
01.3	During changing state: (average) PE decreases AND (average) KE doesn't change ✓ During cooling: (average) KE decreases AND (average) PE doesn't change ✓	2	Allow alternative ways of referring to each time period. Condone reference to seconds rather than minutes. If no other mark awarded, allow 1 mark for a correct description of only the PE during both periods OR of only the KE during both periods OR of a decrease in PE while changing state <u>and</u> a decrease in KE during cooling.
Total		6	

Question	Marking guidance	Mark	Comments
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Relevant statement about plasma, energy extraction, or safety ✓ Relevant discussion ✓	e.g. Achieving/maintaining/containing high temperature plasma; extracting energy from plasma/neutrons; shielding from high energy neutrons	2	AO1 AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Calculates mass difference ✓ Converts their mass difference to energy in joules ✓ 2.6×10^{-13} (J) ✓	For MP1, condone calculation error if correct working seen. Expect 2.87×10^{-30} kg or 1.73×10^{-3} u. MP2: e.g. uses $E = mc^2$ OR uses 931.5 MeV and 1.60×10^{-19} . Allow power of ten error for eV to J conversion. Allow 2.5×10^{-13} (J).	3	AO2 AO2 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Reaction 1 occurs twice because: reaction 2 has to occur twice; or because 2 He-3 are required ✓ Reaction 2 occurs twice because 2 He-3 are required ✓		2	AO3 AO3
Total			7	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Calculates total mass difference OR finds difference between U-238 and U-237 OR adds U-237 mass to neutron mass ✓ Decay can't happen because: total mass increases OR because difference in uranium nuclide masses is less than neutron mass OR mass of U-237 + neutron > mass of U-238 ✓	Expect total mass difference of 0.0061 u or 1.10×10^{-29} kg. Difference in uranium nuclide masses = 1.00206 u. Mass of U-237 + neutron = 238.00698 u.	2	AO2 AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	Basic description of chain reaction ✓ Detail of nuclear process of fission ✓ Detail about chain reaction ✓	MP1: e.g. neutron causes fission of uranium-235 leading to more neutrons. MP2: e.g. neutron absorbed by U-235 nucleus; a U-236 nucleus formed. MP3: e.g. fission produces several neutrons that go on to induce further fissions (of U-235 nuclei).	3	AO1 AO1 AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Water/heavy water OR graphite/carbon ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	(neutrons are) in thermal equilibrium with the fuel/moderator ✓	Allow idea that neutrons have same (kinetic) energy as fuel/moderator (atoms)	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.5	Idea that chain reaction requires number of neutrons per fission to be ≥ 1 ✓ Idea that fission probability depends on KE/speed of neutron ✓	MP2: e.g. faster neutrons are less likely to be absorbed by U-235 nuclei (to induce fission). Condone “won’t be” for “are less likely to be”.	2	AO1 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.6	$2.5 \times 10^{-2} = 2.0 \times 10^6 \times (0.84)^n$ ✓ 105 ✓	Allow 104.	2	AO3 AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.7	Energy to heat solid to melting point: $3.3 \times 10^5 \times 1200 \times (98 - 20) = 3.09 \times 10^{10} \text{ J}$ OR Energy to heat liquid to 560 °C: $3.3 \times 10^5 \times 1300 \times (560 - 98) = 1.98 \times 10^{11} \text{ J} \checkmark$ Energy to melt solid: $3.3 \times 10^5 \times 110000 = 3.63 \times 10^{10} \text{ J} \checkmark$ $2.65 \times 10^{11} \text{ (J)} \checkmark$	MP1: one SHC calculation MP2: latent heat calculation MP3: final value For MP1: do not allow 1250 for c. Value of c must match the state of matter. Allow power of ten errors for MP1 and MP2. $3.65 \times 10^{10} \text{ (J)}$ gains 2 marks	3	AO3 AO2 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.8	Comment about problem related to nature of radiation $\checkmark$ Comment about problem related to half-life $\checkmark$	MP1: e.g. steel/lead/very thick concrete shielding needed for gamma; thermal energy released so needs cooling (in water) MP2: e.g. store for at least 5 half-lives/150 years	2	AO2 AO2
Total			16	

Question	Answers	Additional comments/Guidelines	Mark	AO
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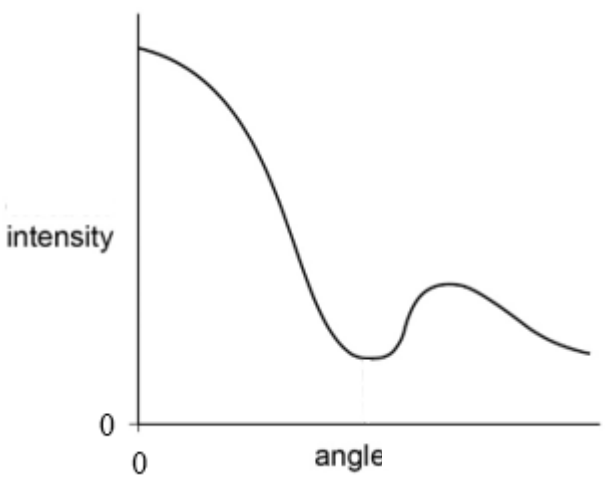
16	B	Nucleons have less potential energy in the nucleus than as separate nucleons.	AO1
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19	C	$^{62}_{28}\text{Ni}$	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Evidence of determination of charge on silicon OR charge on alpha ✓ Convert MeV to J (to find E_k) ✓ Substitution into EPE equation ✓ $6.2 \times 10^{-15} \text{ (m)}$ ✓	$Q_{Si} = 14 \times 1.6 \times 10^{-19} \text{ or } 2.24 \times 10^{-18} \text{ (C)}$ $Q_{\alpha} = 2 \times 1.6 \times 10^{-19} \text{ or } 3.2 \times 10^{-19} \text{ (C)}$ $E_k = 6.5 \times 1.6 \times 10^{-13} \text{ or } 1.04 \times 10^{-12} \text{ (J)}$ E.g. for MP3: $their E_k = \frac{their Q_{Si} \times their Q_{\alpha}}{4\pi\epsilon_0 r}$ Allow POT error in MP1, MP2 and MP3	4	2 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Idea that before reaching the nucleus: the alpha particle is stopped OR all the KE (of the alpha particle) is transferred in EPE ✓	Allow alternative answer suggesting that: value does not take into account radius of alpha particle. OR calculation assumes collision is head-on.	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	Idea that the (change in) EPE is the same (in both situations). ✓ Charge on Y greater (than charge on X) ✓ Therefore, estimated value for Y is greater (than value for X) ✓	Accept formula for EPE. Allow reverse arguments MP3 can be given on its own provided there is some support, e.g. 'Y has more protons (than X).'	3	$2 \times \text{AO1}$ $1 \times \text{AO2}$

Question	Answers	Additional comments/Guidelines	Mark	AO
04.4	Curve correct With central max, one min inflection point and no points of zero intensity Second peak less than half height of central max above min. ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.5	Idea that electron diffraction depends on diameter/size of nucleus OR electron diffraction does not depend on proton number/charge (on nucleus) ✓ Radius the same as nucleon number the same (40) ✓	In MP1 do not accept 'number of nucleons' for diameter. Allow 'mass (number)' for 'nucleon number'.	2	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
04.6	Best-fit line drawn ✓ Point on line read ✓ $\ln R = \frac{1}{3} \ln A + \ln R_0$ seen OR Correct use of e^x for their point ✓ Answer between 1.04 fm and 1.07 fm ✓	Equal scatter of points on either side of line. Expect a line going through (2.7, 0.95) and (3, 1.05). Reject multiple or thick lines. Only allow a plotted point if on the line. Ignore POT error. Only MP1 and MP3 can accessed for using intercept on y-axis = $\ln R_0$. Note $\ln(1.04) = 0.039221$ $\ln(1.07) = 0.06759$	4	AO3
Total			15	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	Idea that only a small amount of the rest energy of the nucleons is lost when the nucleus is formed. OR The rest energy of the nucleons and rest energy of the nucleus are very similar and the binding energy is (only) the difference between them. OR binding energy = mass defect $\times c^2$ AND rest energy = rest mass/mass of nucleus/nucleons $\times c^2$ AND mass defect $\ll$ rest mass/mass of nucleus/nucleons OR Binding energy = rest energy of (individual) nucleons – rest energy of nucleus AND rest energy of nucleons $\approx$ rest energy of nucleus. ✓	Allow reverse in terms of splitting up the nucleus into nucleons Alternative : (binding energy =) Δmc^2 $= m_{\text{nucleons}}c^2 - m_{\text{nucleus}}c^2$ AND $m_{\text{nucleons}}c^2 \approx m_{\text{nucleus}}c^2$ Condone some confusion between rest energies. Condone $<$ for $\ll$	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	Identifies X as neutron ✓ Converts mass of neutron to MeV to give answer that rounds to 940 to 3sf (MeV) ✓	Identification can be by mass used in MP2. Expect to see $1.00867 \text{ (u)} \times 931.5 \text{ (MeV u}^{-1}\text{)}$ = 939.6 (MeV) Allow use of mass of neutron in kg (1.675×10^{-27} to give 942(.2 MeV)) Allow use of 1.67×10^{-27} kg (to give 939.4 (MeV)) only if neutron identified.	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	Q subtracted from binding energy of He-3 ✓ Divided by 2 to get 2.2(2) (MeV) ✓	Allow 1 mark for an answer of 5.5 MeV from adding Q and binding energy of He-3 and halving.	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.4	energy released per (fusion) reaction ✓ pressure / temperature required (for optimum rate of fusion reaction) ✓	'energy released' is insufficient for MP1 For MP2 allow idea of amount of energy required OR strength of magnetic field required	2	AO1
Total			7	

Question	Key	Answer	AO
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12	D	161 MeV	AO2
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13	C	$Am_n + Z(m_p - m_n) - M$	AO1
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14	B	boron graphite	AO1
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18	D	6.6×10^{19}	AO2
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20	B	The determination of the critical mass assumes that the fuel is spherical.	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Uses $E = mc^2$ ✓ Divides by 1.6×10^{-13} to get 934 (MeV) to at least 3 sf ✓		2	1 × AO1 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Attempts to find the mass defect ✓ 12.9 (MeV) ✓	Expect to see $2 \times 3.016029 - 4.002603 - 2 \times 1.007823$ $= 0.0138$ (u) Condone one missing $\times 2$ for mp1 Allow ecf for their answer to 4.1 930 MeV gives 12.8 (MeV) 931.5 MeV also gives 12.9 (MeV)	2	1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	Increase in binding energy (per reaction) $= 1.23489 - 0.35593 (= 0.87896 \text{ (pJ)})$ ✓ Each hydrogen cycle has two second-stage reactions / 3 rd stage requires two 2 nd stage reactions (so total contribution $= 2 \times 0.87896 = 1.75792 \text{ (pJ)} =$ $1.75792 \times 10^{-12} \text{ (J)}$) ✓	Candidates must articulate the reason for the factor of 2 in order to get MP2 Allow ${}^3_2\text{He}$ for 2 nd stage reactions.	2	1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.4	Two protons fusing to produce one deuteron ✓ All details correct: ${}_1^1\text{p} + {}_1^1\text{p} \rightarrow {}_1^2\text{H} + {}_1^0\text{e}^{(+)} + \nu$ ✓	Allow ${}_1^1\text{H}$ for proton. Allow β for positron Penalise antineutrino and/or electron	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.5	MAX 3 for any of the following: ✓✓✓ - Divides mass of hydrogen by mass of 1 mol to get number of moles AND multiplies by Avogadro number to find number of hydrogen nuclei - Dividing by 4 to get number of hydrogen cycles - Multiplies a number of cycles/hydrogen nuclei by 4.5×10^{-12} to get energy released - Divides an energy by 4×10^{26} to get time AND converts to years OR MAX 3 for any of the following: ✓✓✓ - Divides 4×10^{26} by 4.5×10^{-12} to get number of reactions per second - Multiplies the number of reactions per second by 4 to find the number of hydrogen ions used per second - Divides by Avogadro number to find number of moles AND multiplies by molar mass converted per second - Divides 2×10^{30} by mass per second to get time AND converts to years 1.1×10^{11} (years) ✓	If $E = mc^2$ is used allow only 2nd and 4th bullet point Expect to see 1.2×10^{57} Expect to see 3.0×10^{56} Expect to see 5.4×10^{45} (J) Expect to see 3.5×10^{18} (s) Expect to see 8.9×10^{37} Expect to see 3.6×10^{38} Expect to see 5.9×10^{11} (kg) Expect to see 3.4×10^{18} (s) 1.07×10^{11} to 3SF. Allow 1 SF answer. Correct answer gets 4 marks	4	AO2

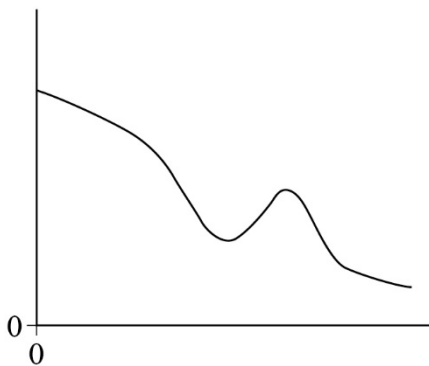
Question	Answers	Additional comments/Guidelines	Mark	AO
04.6	Electrostatic repulsion is greater / greater electrostatic potential energy as each nucleus has a greater charge ✓ ...so plasma will need a higher temperature so that kinetic energy of the nuclei is sufficient ✓		2	1 × AO1 1 × AO2
Total			14	

Question	Answers	Additional comments/Guidelines	Mark	AO
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14	D	140 MeV	AO3
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16	B	7	0.0005		AO2
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17	D	1 and 2 and 3	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Graph showing central maximum, non-zero first minimum and further reduction to a non-zero level. ✓	 Accept the correct presence of further minima.	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Uses $\lambda = \frac{h}{p}$ to attempt to find the de Broglie wavelength of the electrons. ✓ The idea that, (for diffraction,) the (de Broglie) wavelength should be comparable to the size of the nucleus. ✓ Gets an answer of 3.0×10^{-15} (m) ✓	Condone – they are about the same size so suitable Max 2 if no indication about whether it is or is not suitable.	3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	The idea that Pb-208 has a greater/different nuclear diameter than Pb-204 (as $R = R_0 A^{\frac{1}{3}}$) (and that since the technique actually measures diameter), the measured value will be greater/different for Pb-208 than for Pb-204 ✓	Accept statement that difference will be very small because not much difference in nuclear diameter.	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.4	Any TWO from: ✓✓ Converts alpha particle energy to J $E = \frac{Q_1 Q_2}{4\pi\epsilon_0 r}$ seen Selects $2 \times 1.6 \times 10^{-19}$ and $82 \times 1.6 \times 10^{-19}$ as Q_1 and Q_2 Then: answer that rounds to 2.8×10^{-14} (m) ✓	Expect 1.344×10^{-12} J	3	2 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.5	Idea that the distance of closest approach is the same because it depends on the proton/charge number (and not the nucleon number). ✓		1	AO2

Total			9	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Any two from: ✓✓ - calculates volume of nucleus - calculates mass of nucleus - uses their mass and their volume to calculate a density $2.7 \times 10^{17} \text{ (kg m}^{-3}\text{)} \checkmark$	Allow mass determination using 54 u or $26 m_p$ and $28 m_n$	3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Any two from: ✓✓ - determines R_0 using data from 03.2 - uses $R = R_0 A^{1/3}$ to determine their A - uses their $A - 26$ to determine their number of neutrons 32 ✓	Allow use of ratios Correct ratio equation Correct A determined	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Any three from: ✓✓✓ - determines charge on iron nucleus OR charge on alpha particle - uses (change in) electric potential - sets their (change in) electric potential energy equal to initial kinetic energy in some form - determines the mass of an alpha particle from its specific charge. $8.5 \times 10^6 \text{ (m s}^{-1}\text{)} \checkmark$	eg for bullet point 3: $\frac{Q}{4\pi\epsilon_0 r} = \frac{1}{2} \frac{m}{q} v^2 \text{ seen}$	4	1 × AO1 3 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.4	$5(.0) \times 10^{-14} \text{ m}$ Idea that closest approach depends on the charge on the target nucleus AND The charge on both nuclei is the same ✓		1	AO1

Total			11	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Identification of areas for fission and fusion ✓ Description of fission and fusion ✓ Consideration of the difference in binding energy per nucleon at the two locations ✓ Consideration of the impact of the difference in number of nucleons involved, leading to idea that fission releases more energy ✓	Fusion on LHS and fission on RHS Fission: energy released when nucleus splits into smaller nuclei Fusion: energy released when small nuclei combine into larger nuclei BEPN greater on LHS/fusion side Very many more nucleons on RHS/fission side so energy released greater Evidence for MP3 and MP4 may be seen in suitable calculations	4	2 × AO1 2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Any two from: ✓✓ - determines binding energy for one nucleus - determines their energy released from their binding energy calculations and not including neutrons - converts their energy to J $2.6 \times 10^{-11} \text{ (J)}$ ✓		3	2 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	Any two from: ✓✓ - calculates the mass of 26 protons in u or kg - calculates the mass of 30 neutrons in u or kg - subtracts mass of iron nucleus (in u or kg) from their total mass in u - converts their difference into MeV 471 (MeV) ✓	Accept use of alternative units Accept range 470 to 480	3	AO3

Total			10	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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8	D	contain atoms with a small nucleon number.	AO1
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9	A	to decrease the number of thermal neutrons	AO1
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10	D	$4.5 \times 10^{-9} \text{ kg s}^{-1}$	AO2
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Unknown Unknown - Unit 4 - Q01

Question	Marking guidance	Mark	Comments
01.1	substitutes into $E = UA\Delta\theta$ for any part – insulated or not ✓ uses standard values ✓ adds their values for all four parts ✓ 2660 (W) cao but condone sf ✓	4	
01.2	Neutron correct in both places ${}^1_0\text{n}$ ✓ Xenon correct ${}^{144}_{54}\text{Xe}$ ✓	2	
01.3	ANY 2 from: Converts MeV into J ✓ Divides 600×10^6 by candidate's energy value ✓ Multiplies by $\frac{235}{N_A}$ ✓ PLUS $7.3 \times 10^{-3} \text{ (g s}^{-1}\text{)}$ ✓	3	Look for $200 \times 1.6 \times 10^{-19}$ Or 3.2×10^{-11} Key values to look for: $1.88 \times 10^{11} \text{ fissions s}^{-1}$ $3.1 \times 10^{-5} \text{ mol}$
01.4	walls ✓ area × (difference between standard insulation and additional insulated) U -value is recognised or calculated or Q or change in Q is calculated for new values ✓ walls = 36 (432), windows = 32 (403), roof = 21 (252) ✓	3	

Unknown Unknown - Unit 4 - Q01 (continued 2)

01.5	Any three from: • Boron is a strong neutron absorber OR has a high absorption cross-section wtte ✓ • The idea that the magnitude of the neutron flux is proportional to the rate of fission or power ✓ • Inserting more of the control rods into the reactor (absorbs more neutrons) and slows the reaction ✓ • The idea that inserting all of the control rods quickly and completely will do an emergency shutdown ✓	3	
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Question	Marking guidance	Mark	Comments
	Use of $T = I\alpha$ ✓		Use of equation is rearrangement or substitution